



SIMATS
ENGINEERING



SIMATS
Saveetha Institute of Medical And Technical Sciences
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

CSA1402 – Compiler Design

Technical Presentation

Phases of compiler

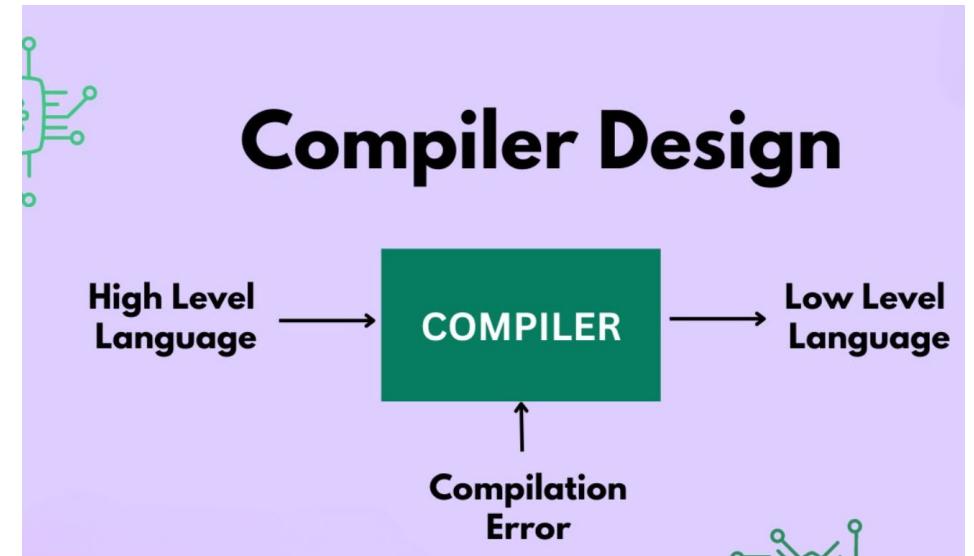
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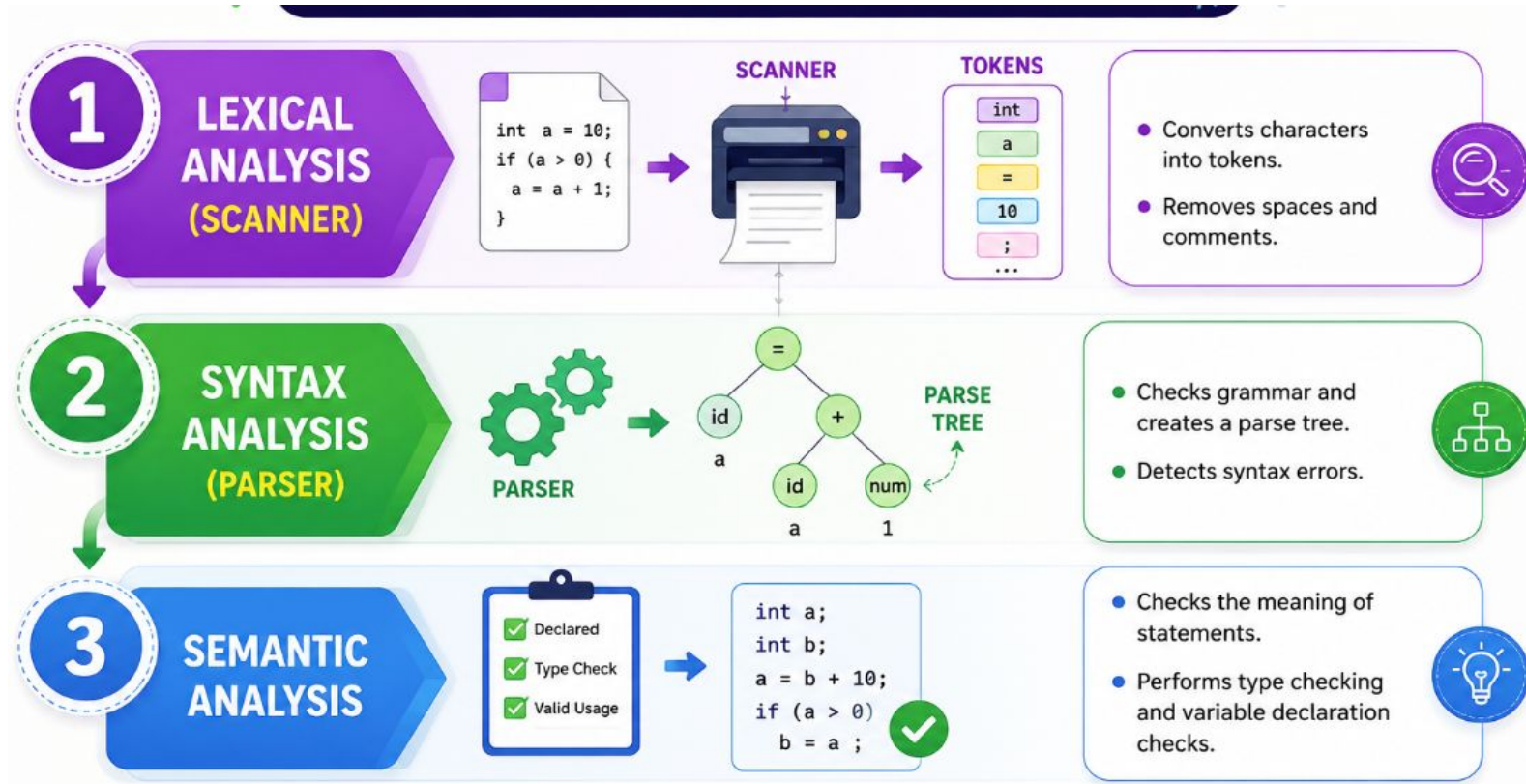
Introduction

What is Compiler?

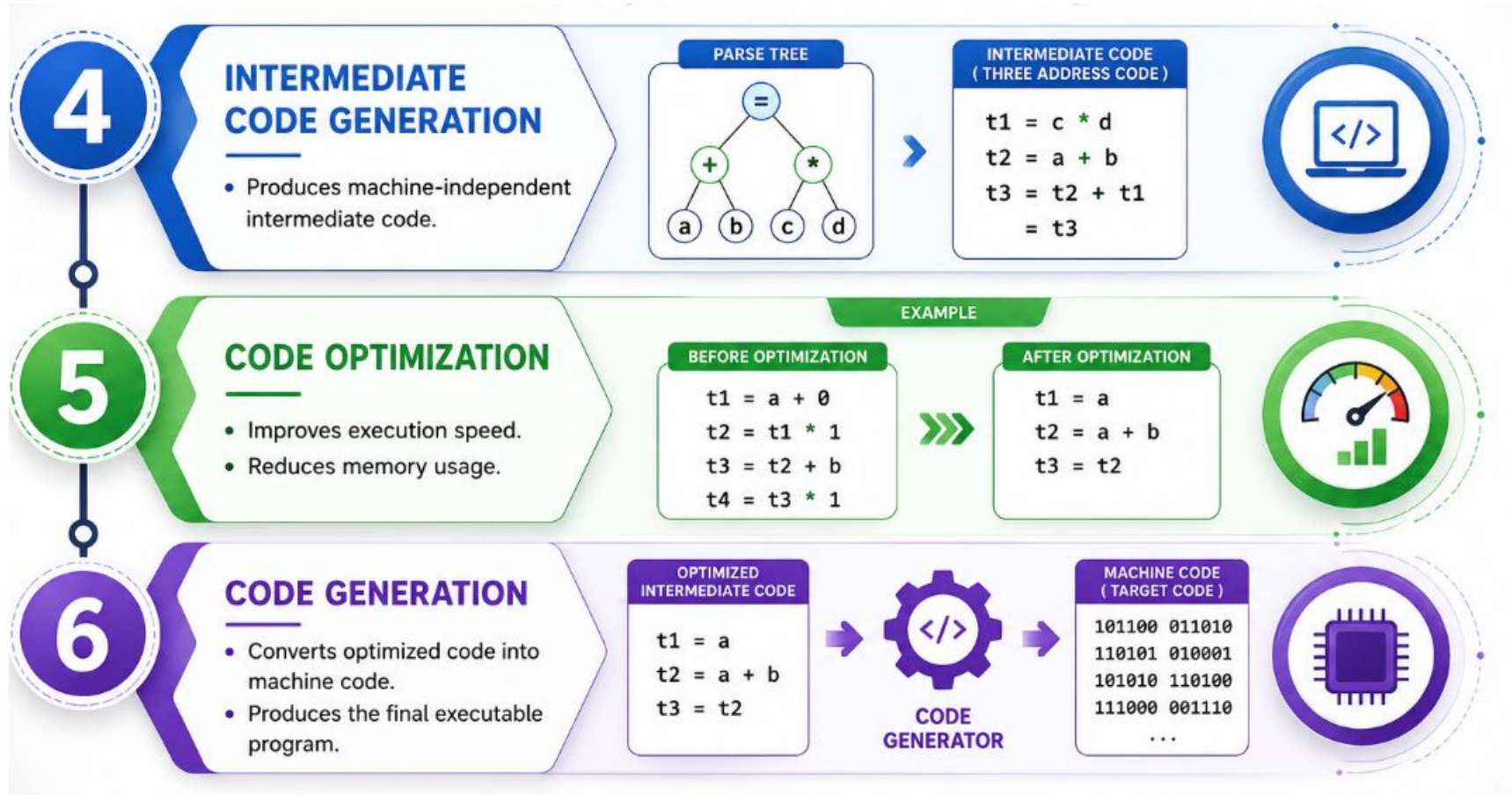
- A compiler is software that translates a high-level programming language into machine code.
- It checks for errors and generates executable code.
- Examples: GCC, Java Compiler (java)



Front End Phases



Back End Phases





Advantages of Compiler Phases



- Detects errors before program execution.
- Improves execution speed through optimization.
- Reduces memory consumption.
- Produces reliable and efficient machine code.
- Simplifies the program translation process.
- Enhances overall software performance.



Conclusion



- A compiler converts source code into machine code through six phases.
- Each phase plays an important role in detecting errors and improving code efficiency.
- The final output is an optimized executable program.



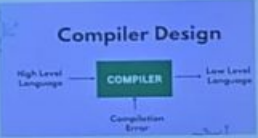
THANK YOU

 **Introduction** 

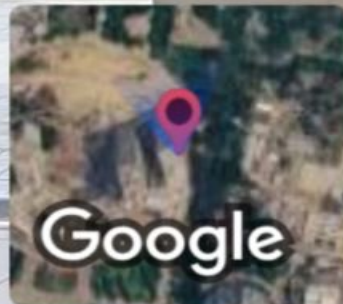
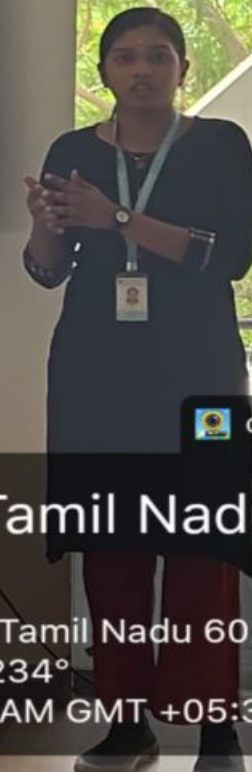
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Compiler Design



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graph LR; A[High Level Language] --> B[COMPILER]; B --> C[Low Level Language]; B --> D[Compilation Error];
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